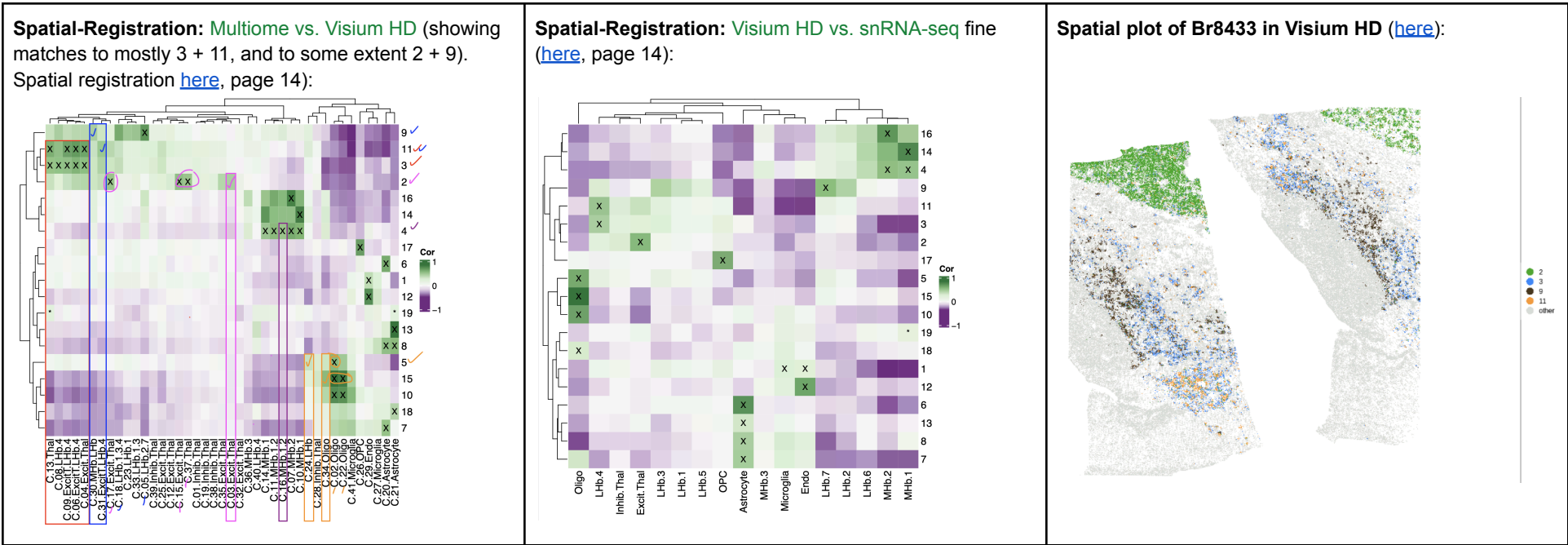


Resolving Habenula Multiome WNN Ambiguous Clusters with Visium-HD

Cynthia SC, Nick Eagles (July 11st, 2025)

The **WNN multiome clusters** below were labeled as **ambiguous** because spatial registration against scRNAseq and/or Visium’3, mean-ratio, and top 50 gene frequency analyses didn’t provide consistent support for a unique cell type. To clarify, we used the higher resolution of the **Habenula Visium-HD** data, together with **spot-plots** depicting the Hb expected region. Below, we summarize the findings and the technical evidence behind our final cell-type assignments.

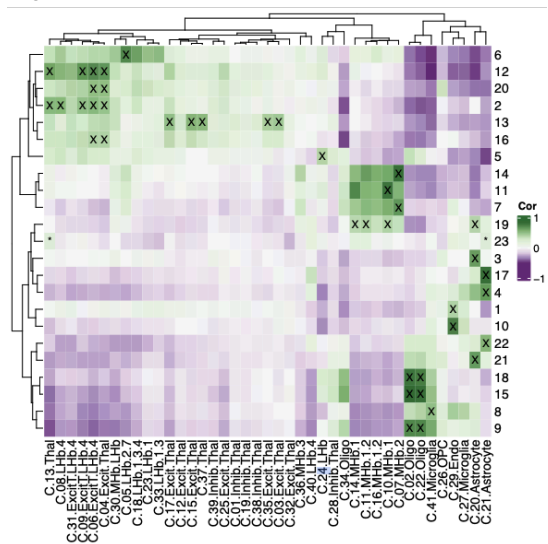
Ambiguous clusters	Cell-type label Visium-DH	VisiumHD support: resolution 1.4
C.03	*Excit.Thal	SpD2
C.04, C.06, C.09, C.13	*LHb.4	SpD3 + SpD11, and some extent to SpD2 + SpD9
C.16	*MHb.1.2	SpD4
C.30	*LHb.7	SpD9
C.31	*LHb.4	SpD11, and some extend to SpD3
C.24	*LHb.4	High gene association with the snRNA-seq reference—DD_LHb (8), DD_MHb (4)—plus strong mean-ratio support for LHb (e.g., C.18 x5, C.40 x2) and clustering with LHb in HClust. But Visium-HD points to Oligos.
C.34	*Oligo	SpD5, SpD10 and SpD15 -- Match with Oligos, present on the MHb(s) HClust clade. Remove from the dataset as it depicts wear celltype associations, likely introducing noise.



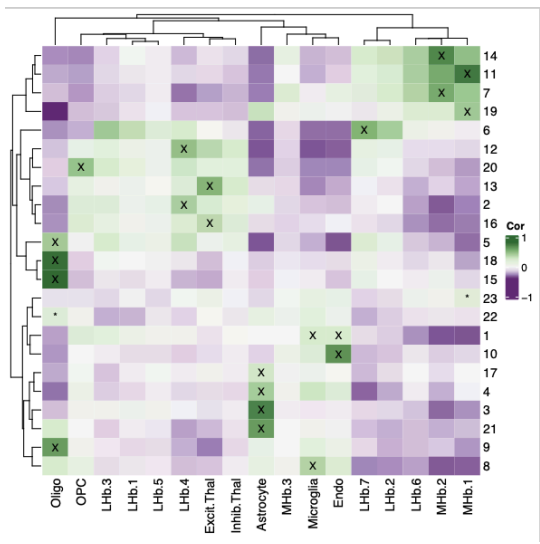
Other Support:

WNN cluster	Cell-type label	Technical support
C.24	*LHb.4	VisiumHD resolution 1.7 for LHb

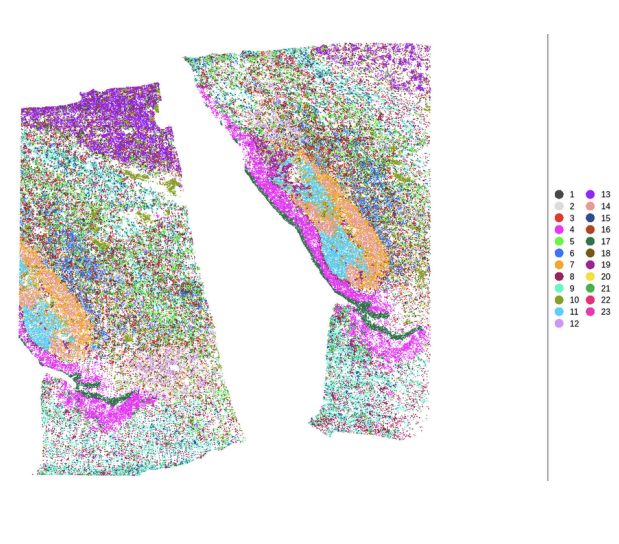
Spatial-Registration: Multiome vs. Visium HD
(showing matches to mostly 5). Spatial registration [here](#),
page 17):



Spatial-Registration: Visium HD vs. snRNA-seq



Spatial plot of Br8433 in Visium HD ([here](#)):



WNN cluster	Cell-type label	Technical support
C.30	*LHb.4	VisiumHD resolution 0.4 for LHb C.30 SpD=02
C.34	*Oligo	C.34 SpD=04

